# Air Quality Sensor Data Analysis

COMP5048 Assignment 2

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## I. Task 1

### A. Data and Preprocessing

We used the Air Quality dataset from the UCI repository [1]. It contains hourly averaged responses of a metal-oxide multisensor device placed at road level in an Italian city, together with reference analyser values and weather readings [2]. The file has 9,357 hourly records from 10 March 2004 18:00 to 4 April 2005 14:00, with no duplicate or missing timestamps, and 13 value columns. Missing values are coded as −200.

Before cleaning, we checked the raw file ourselves. The main findings were:

- −200 is the only missing code. CO(GT) is 18.0% missing, NOx(GT) and NO2(GT) 17.5%, NMHC(GT) 90.2% (valid only until 1 May 2004) and every other column 3.9%.
- Missing values are not random. All sensor columns, C6H6, T, RH and AH go missing together (366 hours), 31 hours have no data at all, NOx and NO2 are missing at 03:00 on 93% of days and CO at 04:00 on 54% of days.
- C6H6(GT) is almost a perfect function of PT08.S2(NMHC) (Spearman 0.99998): the same S2 value always gives the same C6H6 value within 0.1.
- AH matches the water vapour pressure (kPa) calculated from T and RH, so it is a derived column.
- NOx(GT) is in ppb, while NO2(GT) is in µg/m³.
- The sensors drift, as reported in [2]. At a fixed NOx level, the PT08.S3 response ranges from 620 to 923 depending on the month, and PT08.S4 follows T and AH (r = 0.56 and 0.63) more than NO2 (r = 0.16).

Our cleaning script reads the CSV (semicolon separator, decimal comma), drops 114 blank rows and two empty columns, builds one hourly timestamp and replaces −200 with NaN. Nothing is imputed. The script then checks that no rows were lost and no −200 remains, that the number of NaNs in each column equals the number of −200 values in the raw file, and that all other values are unchanged.

### B. Attribute Selection

We selected the timestamp, CO(GT), C6H6(GT), NOx(GT), NO2(GT), PT08.S1(CO), PT08.S5(O3), T and RH (nine attributes) and used them in every task. This covers true concentrations, sensor responses and the environment. We left out NMHC(GT) because it is mostly missing, AH because it is derived from T and RH, PT08.S2 because it duplicates C6H6, and PT08.S3 and PT08.S4 because they showed the strongest drift and temperature effects. Views that show all attributes at once need complete hours, which gives 6,941 hours (74%). October 2004 is the weakest month, at 50%.

Some attributes overlap. C6H6 and CO are strongly related (r = 0.93), and S1 and S5 both follow CO (r = 0.88 and 0.85). A group defined by high CO, C6H6, S1 and S5 is therefore one traffic signal seen several times, not four separate factors, and we describe the groups with this in mind.

### C. Visual Analytic Tool

We built our own web application with four linked views. All views use the same 6,941 complete hours, and the hour of day, weekday, month and season are derived from the timestamp.

- Parallel coordinates [3] is the main view. Each hour is one line, so hours with similar profiles form bundles, and the axes a bundle passes through show which attributes define it. Lines can be coloured by season, time of day (four 6-hour bands), weekday/weekend or saved group.
- A scatterplot of any two attributes shows separation and density for a pair.
- A daily heatmap, with one row per day, shows the daily median of each attribute by colour lightness and cell width, so groups of days stand out.
- A group panel shows, for each hour of day, weekday and month, the share of hours in that slot that belong to a group, together with the group medians.

The axes are ordered analysers, then sensors, then T and RH, so related pollutants sit next to each other and bundles appear as parallel lines. Axes can be dragged to reorder them. Each axis spans the 1st to 99th percentile with its raw end values labelled, because with the full range a few extreme hours pushed most lines to the bottom. The ranges stay fixed when filtering, so views can be compared. With almost 7,000 lines we used low opacity and a fixed random drawing order, so that no season is always drawn on top, and clicking a legend entry isolates a category. We checked the colours for colour-blind safety; only three passed every pair, so the tool allows three saved groups.

Following “overview first, zoom and filter, then details-on-demand” [4], the user starts from all hours, filters by date, season, month, day type and hour, and then selects hours by brushing ranges on one or more axes, dragging a box in the scatterplot or clicking days in the heatmap. The selection is highlighted in all views and can be saved as a group with its own colour and a thick median line. No clustering algorithm was used; the groups were found only by looking at the views and brushing.

### D. *Discovery of Groupings*

We first coloured the parallel coordinates by season and isolated one season at a time (Fig. 1). Winter lines sit at the bottom of T and the top of RH, and many reach high on NO2. Summer lines meet at the top of T and the bottom of RH, and drop to the floor of the NOx axis. Autumn held most of the lines running along the top of every pollutant and sensor axis. Colouring by time of day showed night lines (00–05) along the bottom of all pollutant axes and afternoon lines (12–17) in the hot, dry bundle.

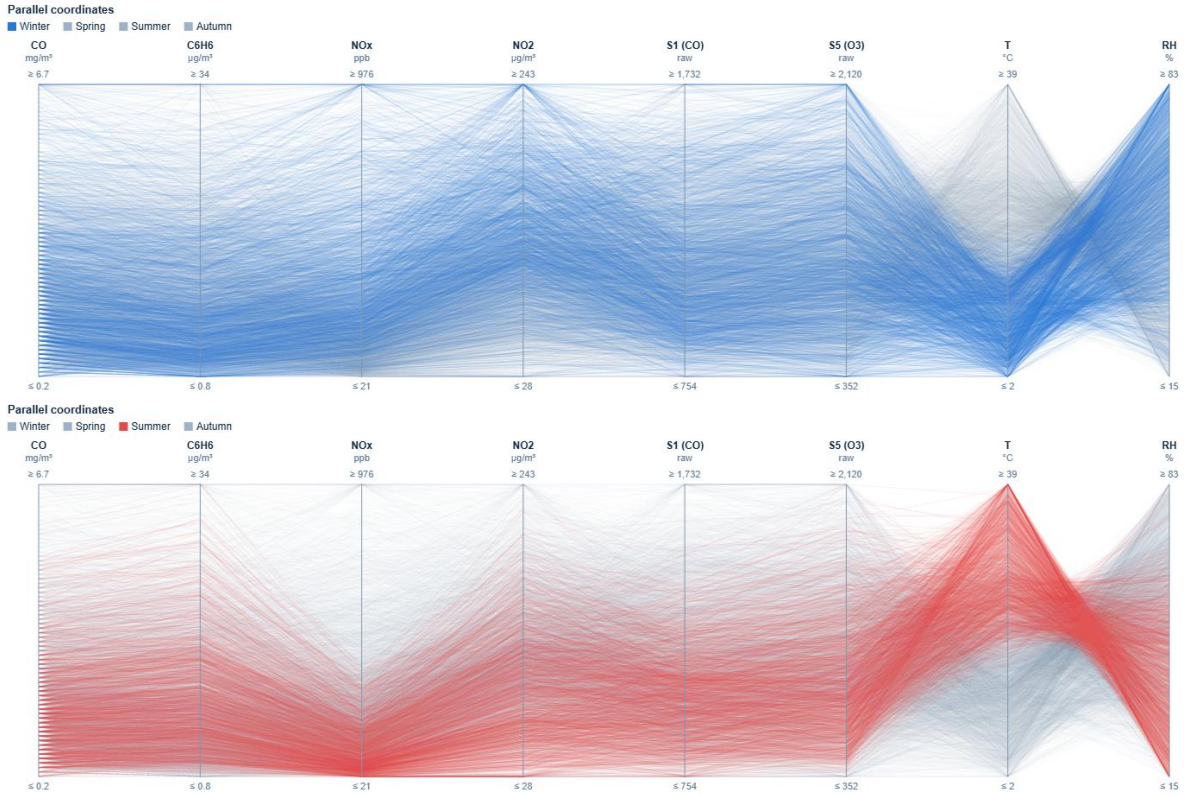



Fig. 1. Parallel coordinates with winter (top) and summer (bottom) isolated.

We then brushed each bundle where it sits on the axes:

- Group A: the upper half of the CO axis (CO ≥ 3.5 mg/m³), where lines also stay high on the other pollutant and sensor axes;
- Group B: the bottom band of CO and NOx (CO ≤ 0.9 mg/m³ and NOx ≤ 90 ppb), where the night lines are;
- Group C: the top third of T and the bottom of RH (T ≥ 26 °C and RH ≤ 35%);
- Group D: the lower part of T (T ≤ 10 °C).

A, B and C were saved as groups (Fig. 2), and D was kept as the current selection because the tool allows three saved groups. Table I summarises the groups using the group panel (Fig. 3). The percentages are the bar heights in that panel; for example, 36% at 08:00 means that 108 of the 303 complete hours at 08:00 belong to Group A.

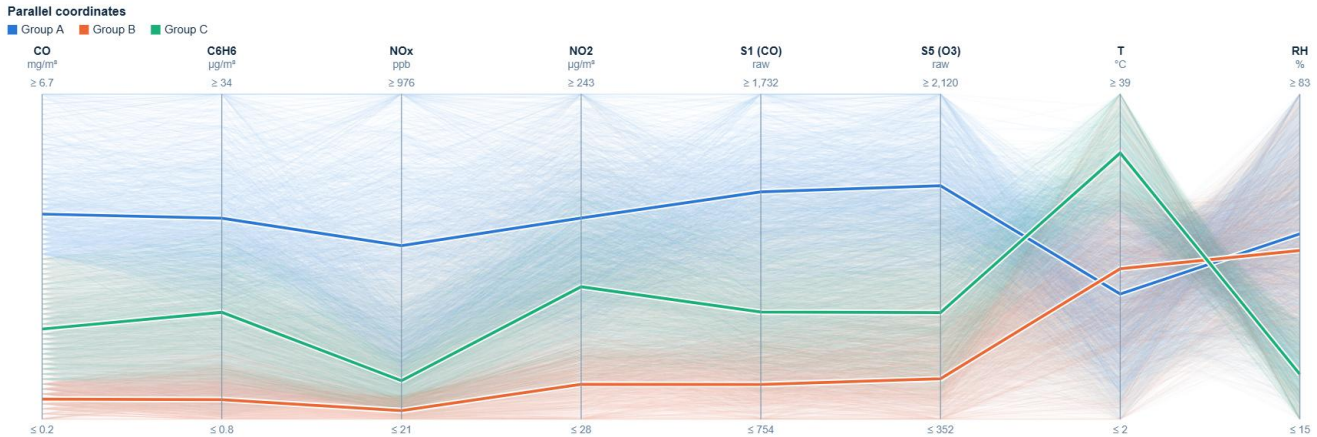



Fig. 2. Parallel coordinates coloured by saved group. Thick lines are the group medians.

TABLE I. GROUPS FOUND IN THE PARALLEL COORDINATES

| Group | Brush | Hours | Defining attributes | Time periods |
|---|---|---|---|---|
| A: traffic peaks | CO ≥ 3.5 | 1,172 (17%) | High on all pollutant and sensor axes. Medians CO 4.3, C6H6 22, NOx 531, NO2 161 (all hours: 1.9, 8.8, 186, 110). T and RH typical | 08–09 (36–37%) and 18–20 (44–51%). Tue–Fri 22–24%, Mon 14%, Sat 9%, Sun 4%. None in Aug 2004; highest Oct–Nov 2004 (29–31%) |
| B: clean nights | CO ≤ 0.9, NOx ≤ 90 | 855 (12%) | Floor of all pollutant and sensor axes. Medians CO 0.6, C6H6 2.8, NOx 46, NO2 51 | 02–06 (64% at 04–05). Sun 24% vs 8–14% on other days. Mostly Apr–Aug 2004 (16–29%); rare Oct 2004–Mar 2005 (4–7%) |
| C: hot, dry afternoons | T ≥ 26, RH ≤ 35 | 906 (13%) | High T (median 33 °C), low RH (24%) and low NOx (134); CO and NO2 typical | 10:00–20:00 (28–32% at 12–17). All days 11–16%. May–Sep 2004, peak Jul 2004 (53%); none Oct 2004–Feb 2005 |
| D: cold, humid hours | T ≤ 10 | 1,431 (21%) | Low T (6.6 °C), higher RH (57%), high NOx (254) and NO2 (131) but CO below typical (1.6) | Mostly 00–09 (25–34%, except 03:00) vs 9–11% at 12–17. All days similar. Mainly Nov 2004–Mar 2005, peak Feb 2005 (78%) |

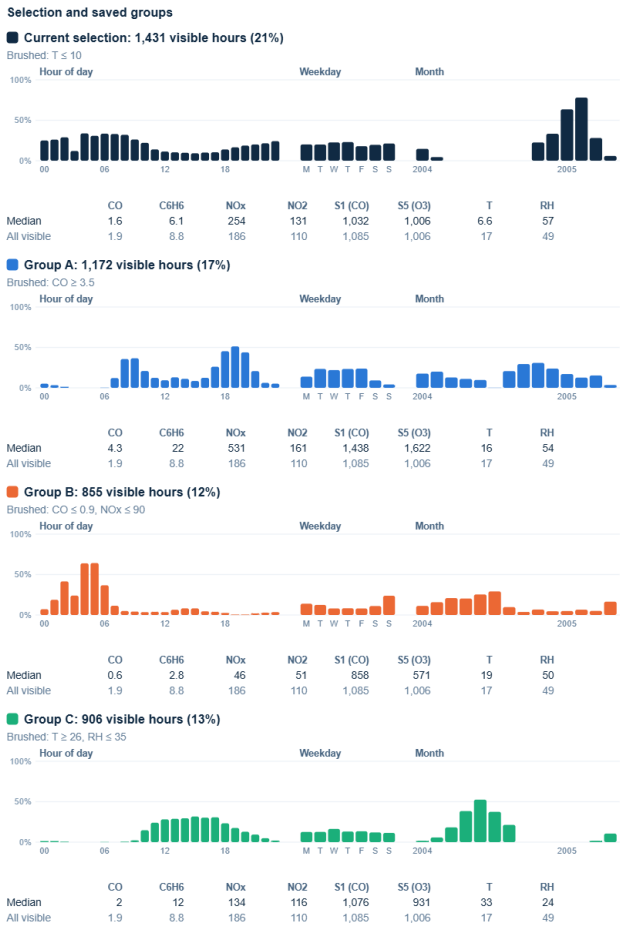



Fig. 3. Group panel: share of hours in each hour of day, weekday and month that belong to each group, with medians. The top card is the cold-hours selection (Group D).

## E. *Groupings over Hours and Days*

The groups depend on the hour and the day, and the same hour can belong to different groups. Afternoons are often in Group C in summer but never in winter, and early mornings are often in Group B in spring and summer but rarely in winter (Fig. 3). The daily heatmap also shows groups of days (Fig. 5). In November 2004 the Sundays (7, 14 and 21 November) are lighter on every pollutant column and have almost no Group A hours, while 22–27 November forms a dark block that looks like a pollution episode lasting several days.

In the scatterplot (Fig. 4), Groups A and B sit at opposite ends of the CO–C6H6 band, while Group C sits on its upper edge, so hot hours have more C6H6 for the same CO. Because C6H6 comes from a temperature-sensitive sensor, we think this is at least partly a sensor effect.

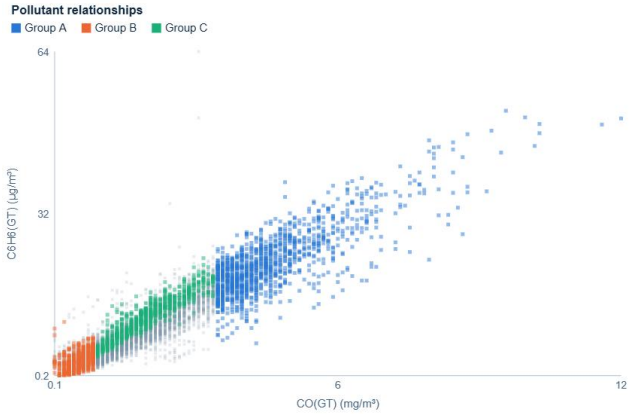



Fig. 4. CO against C6H6, coloured by group.

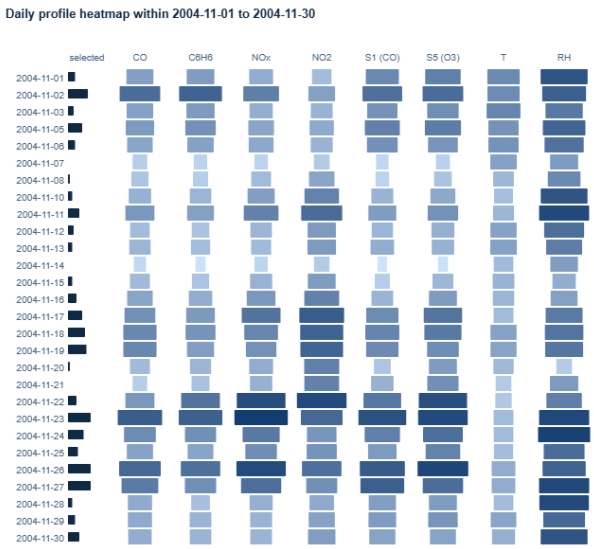



Fig. 5. Daily heatmap for November 2004 with the CO ≥ 3.5 brush active. The left bar counts high-CO hours per day.

## F. *Robustness of the Brushes*

The brush positions are our visual choices, not exact boundaries, because the data is continuous. We moved each brush in both directions and read the group panel again (Table II). The group sizes and percentages change, but the patterns stay the same. A always has the morning and evening peaks and almost no hours at night, on Sundays or in August; B is always night-time and more common on Sundays; C is always a summer daytime group; and D is always a winter group concentrated at night and in the morning.

TABLE II. EFFECT OF MOVING EACH BRUSH (LOWER / USED / HIGHER)

| Group | Brush | Hours | Share of hours in selected slots |
|---|---|---|---|
| A | CO ≥ 3.0 / 3.5 / 4.0 | 1,692 / 1,172 / 814 | 08:00: 49 / 36 / 26%; 19:00: 63 / 51 / 41%; Sunday: 7 / 4 / 3%; 04:00 and Aug 2004: 0% in all |
| B | CO ≤ 0.7, NOx ≤ 70 / 0.9, 90 / 1.1, 110 | 553 / 855 / 1,231 | 04:00: 51 / 64 / 74%; 19:00: 0 / 1 / 3%; Sunday: 16 / 24 / 35% vs Wednesday: 5 / 8 / 11% |
| C | T ≥ 24, RH ≤ 40 / 26, 35 / 28, 30 | 1,221 / 906 / 646 | 14:00: 36 / 30 / 24%; 04:00: 1 / 0 / 0%; Jul 2004: 64 / 53 / 42%; Nov 2004 and Feb 2005: 0% in all |
| D | T ≤ 8 / 10 / 12 | 999 / 1,431 / 1,989 | 04:00: 28 / 34 / 47% vs 14:00: 5 / 10 / 16%; Feb 2005: 61 / 78 / 90%; Jul 2004: 0% in all |

### *G. Limitations*

Groups defined by high CO and C6H6 reflect one traffic signal, and C6H6 depends on a temperature-sensitive sensor. The sensors also drift, so seasonal differences on the S1 and S5 axes are partly instrumental. The groups can overlap: A and C share 102 hours and B and C share 97, while A and B do not overlap. Finally, because only complete hours are used, months with analyser gaps such as October 2004 have fewer hours, and the 03:00 slot has very few complete hours.

## II. TASK 2

[Temporal analysis to be added.]

## III. TASK 3

[Critical moment analysis and what-if scenarios to be added.]

## IV. TASK 4

[Evaluation of the visual analytic tool to be added.]

## REFERENCES


[1] S. De Vito, “Air Quality,” UCI Machine Learning Repository, 2008, doi: 10.24432/C59K5F.

[2] S. De Vito, E. Massera, M. Piga, L. Martinotto, and G. Di Francia, “On field calibration of an electronic nose for benzene estimation in an urban pollution monitoring scenario,” *Sensors and Actuators B: Chemical*, vol. 129, no. 2, 2008.

[3] A. Inselberg, “The plane with parallel coordinates,” *The Visual Computer*, vol. 1, no. 2, pp. 69–91, 1985.

[4] B. Shneiderman, “The eyes have it: A task by data type taxonomy for information visualizations,” in *Proc. IEEE Symp. Visual Languages*, 1996, pp. 336–343.